NLP machine translation exercises

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Q1 Perform beam search with $k = 2$ on the language model search path in fig. 1. What are the two most probable strings?

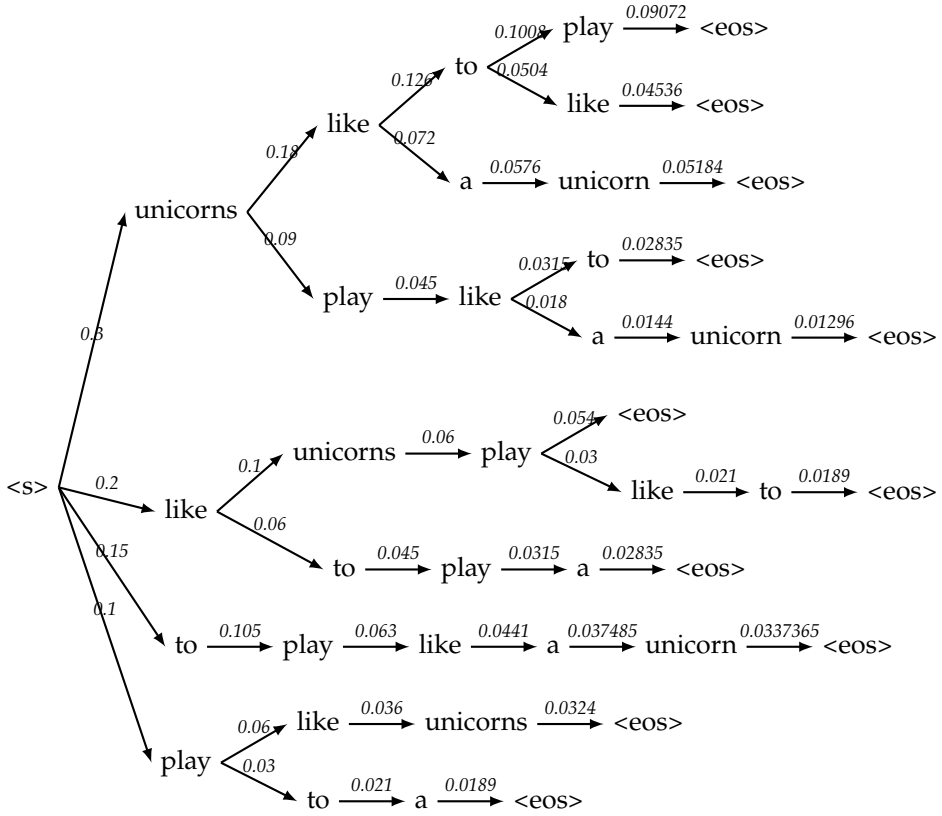


Figure 1: Incomplete search path of an imaginary language model.

Also try with $k = 3$, is the best hypothesis still the same?

🔑 **Q1** The most probable sentences are (see fig. 2):

1. unicorns like to play
2. unicorns like a unicorn

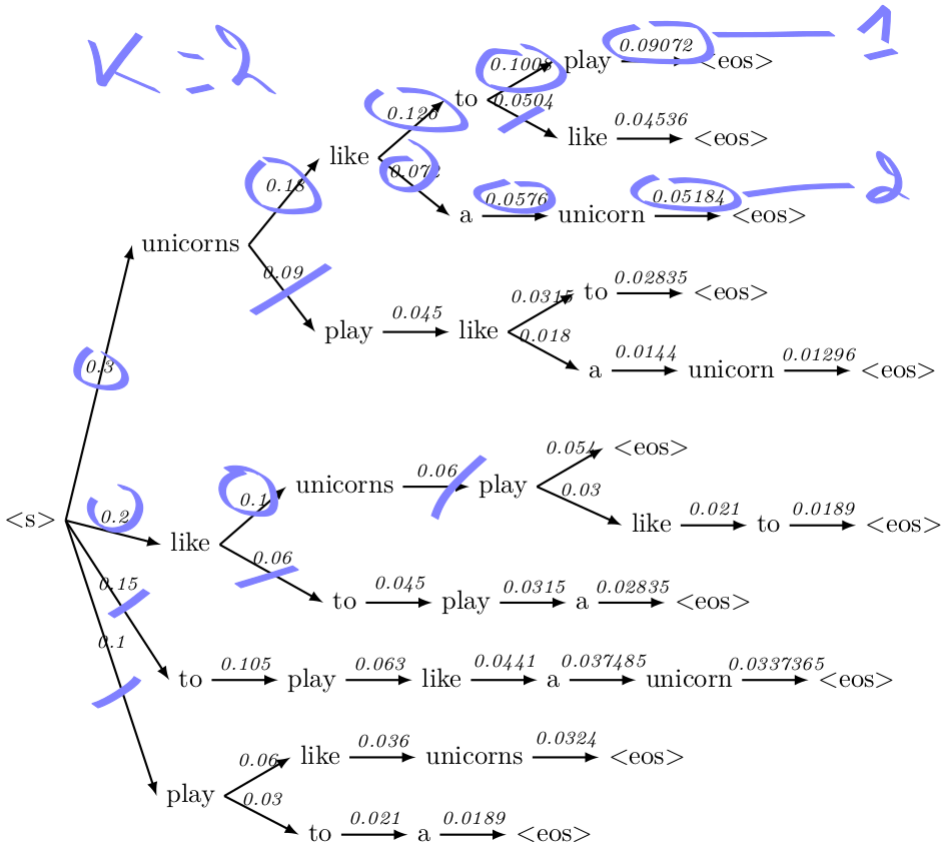


Figure 2: Beam search with $k = 2$: hypotheses considered are circled, hypotheses discarded are barred

Q2 Compute the chrF with $k = \beta = 2$ of the following hypothesis | reference translation pairs:

> **REFERENCE:** `they left together`
>
> **HYPOTHESIS:** `they went away together`

🔑 **Q2**

P 1-gram: $\frac{14}{20} = 0.7$

P 2-gram: $\frac{10}{19} = 0.52$

chrP: $\frac{0.7+0.52}{2} = 0.61$

R 1-gram: $\frac{14}{16} = 0.875$

R 2-gram: $\frac{10}{15} = 0.666$

chrR: $\frac{0.875+0.666}{2} = 0.77$

chrF: $5 \times \frac{0.77 \times 0.61}{4 \times 0.61 + 0.77} = 0.73$